

Medicine by air

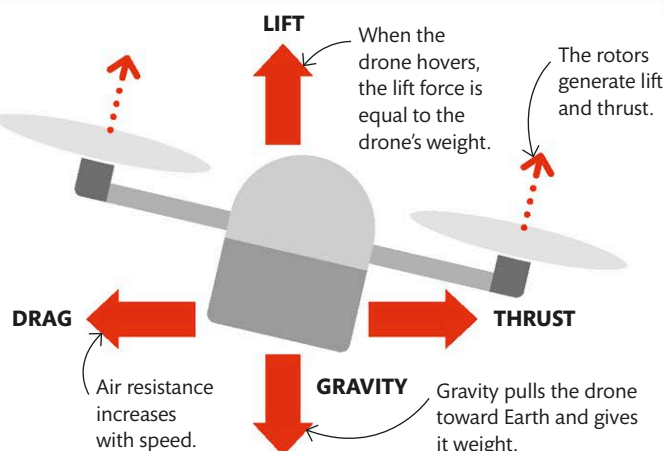
Drones can be used to carry emergency medical packages, such as blood, organs, and personal protective equipment (PPE). Unlike ground-based vehicles, drones do not need roads or railways to reach their destinations. This makes them ideal for making deliveries in hard-to-reach places such as disaster zones.

Each set of propellers provides lift, which makes the drone rise.

The frame provides a steady base for the drone to land on.

FORCES OF FLIGHT

Four forces act on a flying machine—thrust and drag (or air resistance) act horizontally, while gravity and a force called lift act vertically. In a quadcopter drone, four spinning parts called rotors provide the lift and the thrust. When the drone's rotors produce enough lift to counteract gravity, the drone hovers. The drone's operator controls its flight by remotely adjusting the speed of its rotors.



Fighting fires

Drones equipped with high-resolution cameras and other sensors can be used to track the spread of wildfires, helping firefighters plan how to tackle and extinguish the flames. Their small size enables them to reach places that are difficult for aircraft to get to.



Agricultural support

Farmers use drones for tasks such as sowing seeds, spraying fertilizers, or monitoring crop growth and health. The bird's-eye view from a drone can reveal issues that are not obvious from the ground and help farmers gather information over large areas.



Lighting up the sky

Drones equipped with lights can perform spectacular aerial displays. A group of them can be programmed with unique flight paths and other instructions, such as when to switch off and on. Each drone then forms part of a brightly lit, moving 3D image in the night sky.